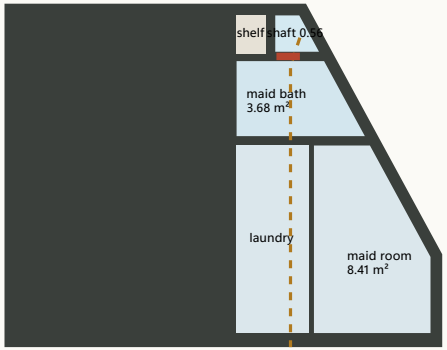
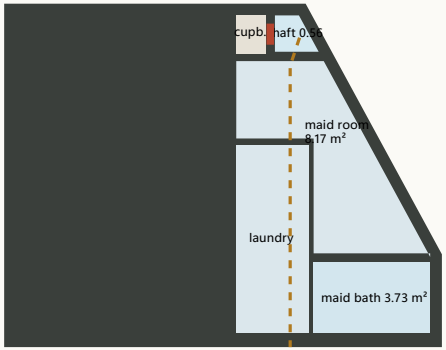
The north-east corner, simplified · a short study (the owner decides once) · Taif / Palm House

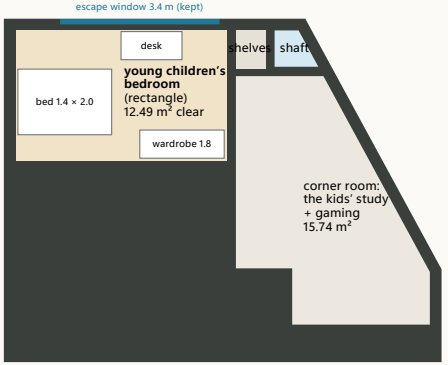
STUDY ONLY, the design files unchanged (V9.3 stays until the owner decides). Above the roof the tower stays as adopted (the 1.25 m square, head (c), 3.5 m). The shaft here is **the corner trapezoid at the cut's tip** north of y 1.0 (its acute corner 61.5°, smooth-rendered); the leftover west of it a shelf cupboard. Clear areas as in the presentation's schedule.



GF (A) · the present arrangement, straight walls



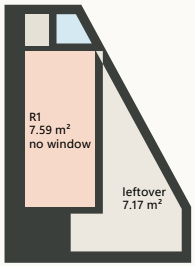
GF (B) · the owner's swap: the maid room in the corner



FF (recommended) · the young children's bedroom in the rectangle

1 · Upstairs: why a good rectangle does not fit the corner

- The cut wall runs **6.1 m**, the full depth of bedroom 1 (y 0 → 5.2), not a small corner. Any square-cornered room in the zone (x 8.95 → the cut, the family link below) touches the cut with one corner only; the largest is **8.87 m² on axes** (≈ 2.2 × 4.0, at y 1.26). Drawn below with the corner shaft: **R1 7.59 m² clear, 1.85 m wide**, and **no outer wall: no window at all** (its walls face the shaft, the kids' room, the family link and a ≈ 7.17 m² leftover along the cut). It fails the rule's "good size" and the escape window.
- **The way to keep the rule:** give the young children **the kids' room's rectangle** (4.7 × 3.0 on axes, **12.49 m² clear**, its 3.4 m north window the escape window: a bed 1.4 × 2.0, a 1.8 m wardrobe, a desk), and turn the corner room into **the older children's study + gaming room** (15.74 m² clear, its two windows on the cut; the slanted outer wall is no problem for a study). **The price:** the young children sleep ≈ 9 m from the suite's lobby instead of ≈ 1 m (along the family living); the master suite is unchanged.
- **If the owner keeps the young children in the corner:** the corner room as above (15.74 m², square inner walls, the cut its window wall) or V9.3's room; neither is a rectangle.



FF · R1: the rectangle in the corner (rejected)

The comparison and the recommendation

Option	Young children's bedroom	Maid room	Maid bath	Shaft (clear) · cleaning	Plumbing · breeze channel	Reading
V9.3 (today)	○ 15.10 m², a wall parallel to the cut inside (not a rectangle)	8.41 m²	3.04 m² + a store 0.56	0.90 m² slim shaft · the panel from the maid bath	the bath's drains must keep off the channel	what the owner wants to replace
FF R1 (the rectangle in the corner)	○ 7.59 m², 1.85 m wide, **no window**	–	–	the corner trapezoid	–	rejected: too small, no window
GF (A) + FF recommended	● 12.49 m² rectangle (the kids' room), escape window kept	8.41 m² (unchanged)	3.68 m² on the cut (its vent window kept) + a shelf 0.52	● 0.56 m² · the panel from the maid bath (a service room)	◐ east cluster; the bath **over** the channel: its WC / shower drains off the line	simple; the constraint stays
GF (B) + FF recommended	● 12.49 m² rectangle (the kids' room), escape window kept	8.17 m² (in the corner, the escape window on the cut kept)	3.73 m² **rectangle** on the kitchen wall, a window in the east wall	◐ 0.56 m² · the panel through the room's cupboard	● the bath beside the east stack, **off the channel**; the channel under the maid room's floor (no cover in the room)	**the cleanest plumbing and a straight bath**

● good · ◐ acceptable with a condition · ○ fails the rule. The shaft in every new option is the corner trapezoid (0.96 m² on axes, 0.56 m² clear, the acute corner 61.5°); with the HRV compartment kept it grows west to x 9.336 (0.9 m² clear) and the shelf shrinks to 0.15 m². Cleaning as adopted: the bird-mesh frames and the top damper from the 0.50 m roof hatch in the tower's south face; the sump from the 0.40 × 0.40 panel at the shaft's foot.

Recommendation

- **Upstairs:** the young children's bedroom in **the kids' room's rectangle** (12.49 m² clear, square corners, the 3.4 m north escape window; a bed 1.4 × 2.0, a 1.8 m wardrobe, a desk); the corner room becomes **the older children's study + gaming room** (15.74 m² clear, its two cut windows kept, square inner walls, the shelf cupboard in the leftover). The only arrangement that gives the young children a clean rectangle of a good size; the corner itself cannot.
- **The shaft: 0.54 m², breeze only** (the corner trapezoid, x 9.775 → the cut, y 0–1.0), the ground floor's two HRV risers **down the west shaft** (it serves the ground floor's middle better; ≈ 10 m of insulated duct inside the attic from the plant room, the west shaft's space to confirm with the MEP engineer). Better than 0.90: a smaller, purely aerodynamic shaft and a real 0.6 m shelf cupboard; the intake stays in the tower's head.
- **Downstairs: (B), the owner's swap.** The maid bath becomes a plain rectangle on the kitchen wall beside the east stack, clear of the breeze channel (no drain constraint any more); the maid room keeps the corner, its escape window on the cut and the same size; the laundry unchanged. Its only weak point: the shaft's access panel opens inside the maid room's cupboard (no service room touches the shaft in (B)); a lockable panel, used twice a year.
- **If the owner wants the young children next to the suite:** keep the corner room for them with square inner walls (15.74 m²) and accept the cut as its window wall; it is not a rectangle.

ملخص

النتيجة: جدار القطع طوله نحو 6.1 م ويمتد بطول غرفة النوم 1 كلها، فأكبر غرفة مستطيلة تتسع في الركن نحو 8.9 م² على المحاور (عرضها 1.85 م صافيا) وبلا أي نافذة؛ فلا تصلح.

التوصية: غرفة الأطفال الصغار في غرفة الأطفال الحالية المستطيلة (12.5 م² صافيا، بنافذة هروب شمالية بعرض 3.4 م)، والركن يصبح غرفة دراسة وألعاب للأولاد الأكبر (15.7 م²). مجرى الملقف شبه منحرف في رأس الركن بمساحة 0.54 م² للنسيم فقط، وأنابيب التهوية للدور الأرضي تنزل في الشفت الغربي، والباقي رف خزانة.

الدور الأرضي: الخيار (ب) فكرة المالك: غرفة الخادمة في الركن بنافذة الهروب، وحمامها مستطيل على جدار المطبخ بجوار المجرى الشرقي وبعيدا عن مجرى النسيم. الثمن الوحيد: فتحة صيانة المجرى من داخل خزانة غرفة الخادمة.

Study checks **passed** (5) · sources: data/rooms_gf.csv / rooms_ff.csv (V9.3), studies/malqaf_tower.typ, studies/air_system.typ. Concept only; every size assumed.